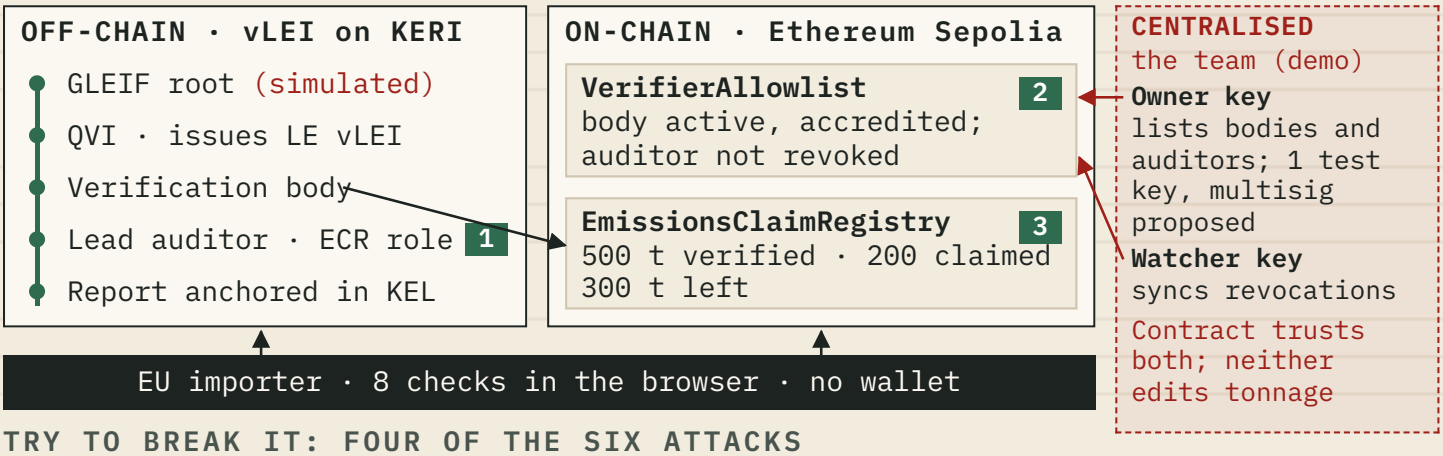
A carbon number is only as trustworthy as the person who signed it.

1 Who signed?
The lead auditor's vLEI role credential (ECR), checked against a GLEIF-rooted chain (simulated root in the demo).

2 Authorised at the registering block?
At the block that registers the report, the contract checks that the body is active and accredited and the auditor is not revoked.

3 Tonnes claimed once?
Within this deployment: the ledger deducts each shipment from the 500 t verified and rejects a batch claimed twice or a claim beyond what remains.

ARCHITECTURE



TRY TO BREAK IT: FOUR OF THE SIX ATTACKS

1. **Tampered value** (1.8 edited to 1.2 tCO2e/t): **check 2 fails, DISCLOSURE_TAMPERED**; no contract is called.
2. **Tonnes claimed twice**, or 400 t when 300 t are left: **reverts BatchAlreadyClaimed / ExceedsVerifiedTonnage** (dry run).
3. **Auditor who left** (ECR revoked, watcher syncs): next report **reverts AuditorNotAuthorized on Sepolia**, status 0 · tx 0xa8b6b8a8…fc5
4. **Impostor body** listed with a stolen owner key (simulated): **contract accepts, status 1**; **check 7 rejects, wrong root** · tx 0x295abe6c…013

REAL SCALE

EU imports of CN 7318 from Taiwan, 2025 (Eurostat): **371,270 t**. Each 1% declared on a verified value instead of the 2026 default moves about **11,060 tCO2e** of declaration onto verifiers' signatures.

MEASURED

1,080 tests; fuzz, Slither, bounded Halmos + Maude model check in CI

412/412 non-equivalent mutants killed

19/19 · 0/28 · 4/4 tampered detected · valid rejected · borderline flagged, on inputs we wrote

524,065 gas: one report + one claim (Sepolia gas has no value)

WHAT IT IS NOT

- × Not CBAM compliance
- × Not a physical reduction
- × Simulated GLEIF root
- × Sepolia testnet only
- × Fictional companies, people and LEIs

TRY IT · SCAN FOR THE DEMO

Demo, read-only, no wallet: **https://zuemen.github.io/carbon-lei/**
Repository: **https://github.com/zuemen/carbon-lei**
Real: Sepolia contracts, transactions, verifier, KERI signatures.